Supplementary File of an Efficient GPU-accelerated Dual Halpern Algorithm for Large-scale Security Constrained Economic Dispatch of Renewable Power Systems

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Appendix B: Proof of Theorem 1

Solving the primal formulation and its restricted Wolfe dual can be equivalently characterized through the following monotone inclusion. Specifically, one needs to find a point $s^* \in \mathbb{R}^m \times \mathcal{W} \times \mathbb{R}^n \times \mathbb{R}^n$ satisfying $0 \in \mathcal{T}s^*$, where the maximal monotone operator $\mathcal{T}$ is defined as

$$\mathcal{T}s = \begin{pmatrix} -\partial\delta^*_{\mathcal{B}}(-y) + Ax \\ Qw - Qx \\ -\partial\delta^*_{\mathcal{D}}(-z) + x \\ c - A^*y + Qw - z \end{pmatrix}, \quad \forall s = (y, w, z, x) \in \mathbb{R}^m \times \mathcal{W} \times \mathbb{R}^n \times \mathbb{R}^n. \tag{1}$$

For the self-adjoint positive semidefinite operator $\mathcal{H}$, by utilizing the factorization result for degenerate preconditioners, there exists a bounded and injective linear operator $\mathcal{C}$ such that $\mathcal{H} = \mathcal{C}\mathcal{C}^*$, where $\mathcal{C}^*$ denotes the adjoint of $\mathcal{C}$ [1]. Based on $\mathcal{H}$ and the monotone operator $\mathcal{T}$ in (1), define

$$\begin{aligned} \widetilde{\mathcal{T}} &= \mathcal{C}^*(\mathcal{H} + \mathcal{T})^{-1}\mathcal{C}, \\ \widetilde{\mathcal{F}} &= 2\widetilde{\mathcal{T}} - \mathcal{I}, \end{aligned} \tag{2}$$

where $\mathcal{I}$ denotes the identity operator.

The following proposition collects the fixed-point and non-expansiveness properties needed for the subsequent complexity analysis.

Proposition 1. *(Proposition 2.5 in [2]) Given the operators $\mathcal{T}$ defined in (1) and $\mathcal{H}$, it follows that the operator $\widetilde{\mathcal{T}}$ defined in (2) is well-defined and firmly-nonexpansive. Moreover, the operator $\widetilde{\mathcal{F}}$ in (2) is non-expansive, and we further have*

$$\mathcal{C}^*\mathcal{T}^{-1}(0) = \mathcal{C}^* \operatorname{Fix} \widehat{\mathcal{T}} = \operatorname{Fix} \widetilde{\mathcal{T}} = \operatorname{Fix} \widetilde{\mathcal{F}}, \tag{3}$$

where the set of fixed points of an operator $\widehat{\mathcal{T}}$ is denoted as Fix $\widehat{\mathcal{T}}$.

To connect Algorithm 3 with a standard Halpern iteration, introduce the projected auxiliary sequences $\{u^k\}$ and $\{\bar{u}^k\}$ as

$$u^k := \mathcal{C}^* s^k \text{ and } \bar{u}^k := \mathcal{C}^* \bar{s}^k, \quad \forall k \geq 0, \tag{4}$$

where $\{s^k\}$ and $\{\bar{s}^k\}$ are generated by Algorithm 3. Applying $\mathcal{C}^*$ to Step 8 of Algorithm 3 yields

$$u^{k+1} = \frac{1}{k+2}u^0 + \frac{k+1}{k+2}\widetilde{\mathcal{F}}u^k, \quad \forall k \geq 0. \tag{5}$$

Proof of Theorem 1: Since (5) is exactly the Halpern iteration applied to $\widetilde{\mathcal{F}}$, [3, Theorem 2.1] gives, for any $u^* \in \operatorname{Fix} \widetilde{\mathcal{F}}$,

$$\left\| u^k - \widetilde{\mathcal{F}}u^k \right\| \leq \frac{2\left\|u^0 - u^*\right\|}{k+1}, \quad \forall k \geq 0. \tag{6}$$

By (3), every fixed point of $\widetilde{\mathcal{F}}$ can be represented through a zero of $\mathcal{T}$. Thus, for any $s^* \in \mathcal{T}^{-1}(0)$, we may set

$$u^* = \mathcal{C}^* s^*. \tag{7}$$

Since $\mathcal{H} = \mathcal{C}\mathcal{C}^*$, the seminorm induced by $\mathcal{H}$ satisfies

$$\|s\|^2_{\mathcal{H}} = \langle s, \mathcal{H}s \rangle = \left\|\mathcal{C}^* s\right\|^2. \tag{8}$$

Let

$$\widehat{\mathcal{F}} := 2(\mathcal{H} + \mathcal{T})^{-1}\mathcal{H} - \mathcal{I}. \tag{9}$$

Then Step 7 of Algorithm 3 can be expressed as

$$\widehat{s}^{k+1} = \widehat{\mathcal{F}} s^k. \tag{10}$$

Moreover, by the construction of the shadow sequence,

$$\mathcal{C}^*\widehat{\mathcal{F}}s^k = \widetilde{\mathcal{F}}\mathcal{C}^* s^k. \tag{11}$$

Combining (6)–(11), we obtain

$$\begin{aligned} \left\|s^k - \widehat{s}^{k+1}\right\|_{\mathcal{H}} &= \left\|\mathcal{C}^* s^k - \mathcal{C}^*\widehat{\mathcal{F}}s^k\right\| \\ &= \left\|u^k - \widetilde{\mathcal{F}}u^k\right\| \\ &\leq \frac{2\left\|\mathcal{C}^* s^0 - \mathcal{C}^* s^*\right\|}{k+1} \\ &= \frac{2\left\|s^0 - s^*\right\|_{\mathcal{H}}}{k+1} = \frac{2R_0}{k+1}, \quad \forall k \geq 0. \end{aligned} \tag{12}$$

Since Algorithm 3 gives

$$\widehat{s}^{k+1} = 2\bar{s}^{k+1} - s^k, \tag{13}$$

It follows that

$$\left\|\bar{s}^{k+1} - s^k\right\|_{\mathcal{H}} \leq \frac{R_0}{k+1}, \quad \forall k \geq 0. \tag{14}$$

This proves the first assertion.

We next derive the residual estimate. Squaring (12) gives

$$\left\|\widehat{s}^{k+1} - s^k\right\|^2_{\mathcal{H}} \leq \frac{4R_0^2}{(k+1)^2}, \quad \forall k \geq 0. \tag{15}$$

Together with (13), this yields

$$\left\|\bar{s}^{k+1} - s^k\right\|^2_{\mathcal{H}} \leq \frac{R_0^2}{(k+1)^2}, \quad \forall k \geq 0. \tag{16}$$

For compact notation, define

$$\begin{aligned} q &:= (y, w), \\ q^k &:= (y^k, w^k), \\ \bar{q}^{k+1} &:= (\bar{y}^{k+1}, \bar{w}^{k+1}), \end{aligned} \tag{17}$$

and

$$A_Q q := A^* y - Qw. \tag{18}$$

By the block structure of $\mathcal{H}$, (16) implies

$$\begin{aligned}&\left\|\bar{q}^{k+1}-q^k\right\|_\Gamma^2\\&\quad+\frac{1}{\rho}\left\|\rho A_Q(\bar{q}^{k+1}-q^k)+(\bar{x}^{k+1}-x^k)\right\|^2\\&\leq\frac{R_0^2}{(k+1)^2},\quad\forall k\geq 0.\end{aligned}\tag{19}$$

Consequently, it follows that

$$\left\|\bar{q}^{k+1}-q^k\right\|_\Gamma\leq\frac{R_0}{k+1},\quad\forall k\geq 0,\tag{20}$$

and

$$\begin{aligned}&\left\|\rho A_Q(\bar{q}^{k+1}-q^k)+(\bar{x}^{k+1}-x^k)\right\|\\&\quad\leq\frac{\sqrt{\rho}R_0}{k+1},\quad\forall k\geq 0.\end{aligned}\tag{21}$$

The x-update in Algorithm 3 gives

$$\bar{x}^{k+1}=x^k+\rho\left(A_Qq^k+\bar{z}^{k+1}-c\right).\tag{22}$$

Therefore, we can obtain

$$\begin{aligned}&\rho A_Q(\bar{q}^{k+1}-q^k)+(\bar{x}^{k+1}-x^k)\\&\quad=\rho\left(A_Q\bar{q}^{k+1}+\bar{z}^{k+1}-c\right).\end{aligned}\tag{23}$$

Define

$$h^{k+1}:=A_Q\bar{q}^{k+1}+\bar{z}^{k+1}-c.\tag{24}$$

It follows from (21) and (23) that

$$\left\|h^{k+1}\right\|\leq\frac{R_0}{\sqrt{\rho}(k+1)},\quad\forall k\geq 0.\tag{25}$$

Let

$$f_1(y):=\delta^*_{\mathcal{B}}(-y),\qquad f_2(z):=\delta^*_{\mathcal{D}}(-z).\tag{26}$$

The residual mapping $\mathcal{K}$ in Theorem 1 can be equivalently written as

$$\mathcal{K}(s)=\begin{pmatrix}y-\mathrm{Prox}_{f_1}(y-Ax)\\Qw-Qx\\z-\mathrm{Prox}_{f_2}(z-x)\\c-A^*y+Qw-z\end{pmatrix}.\tag{27}$$

The first-order optimality condition of the (y,w)-update is

$$\begin{aligned}0\in&\partial\Phi(\bar{q}^{k+1})+A_Q^*\bar{x}^{k+1}\\&+\rho A_Q^*h^{k+1}+\Gamma(\bar{q}^{k+1}-q^k),\end{aligned}\tag{28}$$

where

$$\Phi(y,w):=\delta^*_{\mathcal{B}}(-y)+\frac{1}{2}\langle w,Qw\rangle+\delta_{\mathcal{W}}(w).\tag{29}$$

Using the proximal characterization and nonexpansiveness, we obtain

$$\begin{aligned}&\left\|\begin{pmatrix}\bar{y}^{k+1}-\mathrm{Prox}_{f_1}(\bar{y}^{k+1}-A\bar{x}^{k+1})\\Q\bar{w}^{k+1}-Q\bar{x}^{k+1}\end{pmatrix}\right\|\\&\quad\leq\left\|\rho A_Q^*h^{k+1}+\Gamma(\bar{q}^{k+1}-q^k)\right\|.\end{aligned}\tag{30}$$

Similarly, the z-update gives

$$0\in\partial f_2(\bar{z}^{k+1})+\bar{x}^{k+1},\tag{31}$$

or equivalently,

$$\bar{z}^{k+1}=\mathrm{Prox}_{f_2}(\bar{z}^{k+1}-\bar{x}^{k+1}).\tag{32}$$

Thus,

$$\begin{aligned}\left\|\mathcal{K}(\bar{s}^{k+1})\right\|\leq&\left\|\rho A_Q^*h^{k+1}+\Gamma(\bar{q}^{k+1}-q^k)\right\|\\&+\left\|h^{k+1}\right\|.\end{aligned}\tag{33}$$

Moreover, we have

$$\begin{aligned}&\left\|\rho A_Q^*h^{k+1}+\Gamma(\bar{q}^{k+1}-q^k)\right\|\\&\leq\rho\left\|A_Q^*\right\|\left\|h^{k+1}\right\|+\left\|\sqrt{\Gamma}\right\|\left\|\bar{q}^{k+1}-q^k\right\|_\Gamma.\end{aligned}\tag{34}$$

Combining (20), (25), (33), and (34), we obtain

$$\left\|\mathcal{K}(\bar{s}^{k+1})\right\|\leq\left(\frac{\rho\left\|A_Q^*\right\|+1}{\sqrt{\rho}}+\left\|\sqrt{\Gamma}\right\|\right)\frac{R_0}{k+1},\quad\forall k\geq 0.\tag{35}$$

This proves the second assertion.

It remains to estimate the objective error. Define

$$\begin{aligned}&e(\bar{y}^{k+1},\bar{w}^{k+1},\bar{z}^{k+1})\\&:=\frac{1}{2}\langle\bar{w}^{k+1},Q\bar{w}^{k+1}\rangle+\delta^*_{\mathcal{B}}(-\bar{y}^{k+1})+\delta^*_{\mathcal{D}}(-\bar{z}^{k+1})\\&\quad-\left[\frac{1}{2}\langle w^*,Qw^*\rangle+\delta^*_{\mathcal{B}}(-y^*)+\delta^*_{\mathcal{D}}(-z^*)\right].\end{aligned}\tag{36}$$

Since $s^*=(y^*,w^*,z^*,x^*)$ solves the KKT system, we have

$$Qw^*-Qx^*=0,\quad Ax^*\in\partial\delta^*_{\mathcal{B}}(-y^*),\quad x^*\in\partial\delta^*_{\mathcal{D}}(-z^*),\tag{37}$$

and

$$A_Qq^*+z^*-c=0,\qquad q^*:=(y^*,w^*).\tag{38}$$

By convexity, we have

$$\begin{aligned}&e(\bar{y}^{k+1},\bar{w}^{k+1},\bar{z}^{k+1})\\&\geq\langle Qw^*,\bar{w}^{k+1}-w^*\rangle-\langle Ax^*,\bar{y}^{k+1}-y^*\rangle\\&\quad-\langle x^*,\bar{z}^{k+1}-z^*\rangle.\end{aligned}\tag{39}$$

Using $Qw^*=Qx^*$, we obtain

$$\begin{aligned}&\langle Qw^*,\bar{w}^{k+1}-w^*\rangle-\langle Ax^*,\bar{y}^{k+1}-y^*\rangle\\&\quad-\langle x^*,\bar{z}^{k+1}-z^*\rangle\\&=-\left\langle x^*,A_Q\bar{q}^{k+1}+\bar{z}^{k+1}-c\right\rangle=-\langle x^*,h^{k+1}\rangle.\end{aligned}\tag{40}$$

Hence, it yields

$$\begin{aligned}e(\bar{y}^{k+1},\bar{w}^{k+1},\bar{z}^{k+1})\geq&-\left\|x^*\right\|\left\|h^{k+1}\right\|\\\geq&-\frac{\left\|x^*\right\|}{\sqrt{\rho}}\frac{R_0}{k+1}.\end{aligned}\tag{41}$$

For the upper bound, (28), (31), and convexity yield

$$\begin{aligned}&e(\bar{y}^{k+1},\bar{w}^{k+1},\bar{z}^{k+1})\\&\leq-\left\langle\bar{x}^{k+1},A_Q(\bar{q}^{k+1}-q^*)+\bar{z}^{k+1}-z^*\right\rangle\\&\quad-\left\langle\rho A_Q^*h^{k+1}+\Gamma(\bar{q}^{k+1}-q^k),\bar{q}^{k+1}-q^*\right\rangle.\end{aligned}\tag{42}$$

Since

$$A_Q(\bar{q}^{k+1}-q^*)+\bar{z}^{k+1}-z^*=h^{k+1},\tag{43}$$

we have

$$\begin{aligned} & e(\bar{y}^{k+1}, \bar{w}^{k+1}, \bar{z}^{k+1}) \\ & \leq -\left\langle \bar{x}^{k+1}, h^{k+1} \right\rangle - \rho \left\langle h^{k+1}, A_Q(\bar{q}^{k+1} - q^*) \right\rangle \\ & \quad - \left\langle \Gamma(\bar{q}^{k+1} - q^k), \bar{q}^{k+1} - q^* \right\rangle. \end{aligned} \tag{44}$$

By adding and subtracting x^* in the first two inner-product terms, we obtain

$$\begin{aligned} & -\left\langle \bar{x}^{k+1}, h^{k+1} \right\rangle - \rho \left\langle h^{k+1}, A_Q(\bar{q}^{k+1} - q^*) \right\rangle \\ & = -\left\langle \bar{x}^{k+1} + \rho A_Q(\bar{q}^{k+1} - q^*), h^{k+1} \right\rangle \\ & = -\left\langle x^*, h^{k+1} \right\rangle \\ & \quad - \left\langle \bar{x}^{k+1} - x^* + \rho A_Q(\bar{q}^{k+1} - q^*), h^{k+1} \right\rangle. \end{aligned} \tag{45}$$

Hence, applying the Cauchy–Schwarz inequality to (45) gives

$$\begin{aligned} & -\left\langle \bar{x}^{k+1}, h^{k+1} \right\rangle - \rho \left\langle h^{k+1}, A_Q(\bar{q}^{k+1} - q^*) \right\rangle \\ & \leq \|x^*\| \left\|h^{k+1}\right\| \\ & \quad + \left\|\bar{x}^{k+1} - x^* + \rho A_Q(\bar{q}^{k+1} - q^*)\right\| \left\|h^{k+1}\right\|. \end{aligned} \tag{46}$$

For the last term in (44), we follow the same estimation strategy as in the proof of Theorem 3.7 in [2]. Instead of directly bounding $\bar{q}^{k+1} - q^*$, we decompose it as

$$\bar{q}^{k+1} - q^* = (\bar{q}^{k+1} - q^k) + (q^k - q^*). \tag{47}$$

Therefore, we have

$$\begin{aligned} & -\left\langle \Gamma(\bar{q}^{k+1} - q^k), \bar{q}^{k+1} - q^* \right\rangle \\ & = -\left\|\bar{q}^{k+1} - q^k\right\|_\Gamma^2 \\ & \quad - \left\langle \Gamma(\bar{q}^{k+1} - q^k), q^k - q^* \right\rangle. \end{aligned} \tag{48}$$

Consequently, by the Cauchy–Schwarz inequality in the Γ-seminorm,

$$\begin{aligned} & -\left\langle \Gamma(\bar{q}^{k+1} - q^k), \bar{q}^{k+1} - q^* \right\rangle \\ & \leq \left\|\bar{q}^{k+1} - q^k\right\|_\Gamma^2 \\ & \quad + \left\|\bar{q}^{k+1} - q^k\right\|_\Gamma \left\|q^k - q^*\right\|_\Gamma. \end{aligned} \tag{49}$$

The nonexpansiveness property in the $\mathcal{H}$-seminorm implies

$$\left\|s^k - s^*\right\|_{\mathcal{H}} \leq R_0, \qquad \forall k \geq 0, \tag{50}$$

and

$$\left\|\bar{s}^{k+1} - s^*\right\|_{\mathcal{H}} \leq R_0, \qquad \forall k \geq 0. \tag{51}$$

By the block structure of $\mathcal{H}$, (51) gives

$$\left\|\bar{x}^{k+1} - x^* + \rho A_Q(\bar{q}^{k+1} - q^*)\right\| \leq \sqrt{\rho} R_0. \tag{52}$$

Similarly, (50) implies

$$\left\|q^k - q^*\right\|_\Gamma \leq R_0. \tag{53}$$

Moreover, from the previously established displacement estimates,

$$\begin{aligned} \left\|h^{k+1}\right\| & \leq \frac{R_0}{\sqrt{\rho}(k+1)}, \\ \left\|\bar{q}^{k+1} - q^k\right\|_\Gamma & \leq \frac{R_0}{k+1}. \end{aligned} \tag{54}$$

Substituting (52) and (54) into (46), we obtain

$$\begin{aligned} & -\left\langle \bar{x}^{k+1}, h^{k+1} \right\rangle - \rho \left\langle h^{k+1}, A_Q(\bar{q}^{k+1} - q^*) \right\rangle \\ & \leq \frac{\|x^*\|}{\sqrt{\rho}} \frac{R_0}{k+1} + \frac{R_0^2}{k+1}. \end{aligned} \tag{55}$$

Meanwhile, combining (49), (53), and (54), we have

$$\begin{aligned} & -\left\langle \Gamma(\bar{q}^{k+1} - q^k), \bar{q}^{k+1} - q^* \right\rangle \\ & \leq \left(\frac{R_0}{k+1}\right)^2 + \frac{R_0}{k+1} R_0 \\ & \leq \frac{2R_0^2}{k+1}, \qquad \forall k \geq 0. \end{aligned} \tag{56}$$

Finally, substituting (55) and (56) into (44) yields

$$\begin{aligned} & e(\bar{y}^{k+1}, \bar{w}^{k+1}, \bar{z}^{k+1}) \\ & \leq \frac{\|x^*\|}{\sqrt{\rho}} \frac{R_0}{k+1} + \frac{R_0^2}{k+1} + \frac{2R_0^2}{k+1} \\ & = \left(3R_0 + \frac{\|x^*\|}{\sqrt{\rho}}\right) \frac{R_0}{k+1}. \end{aligned} \tag{57}$$

Therefore, it yields

$$e(\bar{y}^{k+1}, \bar{w}^{k+1}, \bar{z}^{k+1}) \leq \left(3R_0 + \frac{\|x^*\|}{\sqrt{\rho}}\right) \frac{R_0}{k+1}. \tag{58}$$

This completes the proof.

■

Appendix C: The Restart Strategy and the Updating Scheme of ρ

We first define a suitable merit function grounded in the complexity result established in Theorem 1. Specifically, the idealized merit function is given as:

$$R_{r,t} := \left\|s_Q^{r,t} - s^*\right\|_{\mathcal{H}} \quad \forall r \geq 0, t \geq 0,$$

where $s_Q^{r,t} = (y^{r,t}, w_Q^{r,t}, z^{r,t}, x^{r,t})$ and $s^* = (y^*, w^*, z^*, x^*)$. s^* denotes any solution to the following KKT system.

$$\begin{aligned} & Qw^* - Qx^* = 0, \\ & Ax^* \in \partial \delta_{\mathcal{B}}^*(-y^*), \\ & x^* \in \partial \delta_{\mathcal{D}}^*(-z^*), \\ & -Qw^* + A^*y^* + z^* - c = 0. \end{aligned} \tag{59}$$

Note that $R_{r,0}$ corresponds to the upper bound given by the complexity result in Theorem 1 at the start of the r-th outer iteration. Since s^* is unknown in practice, we approximate $R_{r,t}$ using the computable surrogate:

$$\widetilde{R}_{r,t} := \left\|s_Q^{r,t} - \bar{s}_Q^{r,t+1}\right\|_{\mathcal{H}}.$$

Based on this surrogate merit function, we introduce the following adaptive restart criteria for the sGS-dHPR-ED method:

1. Sufficient decay:

$$\widetilde{R}_{r,t+1} \leq \alpha_1 \widetilde{R}_{r,0};$$

2. Insufficient local progress despite overall decay:

$$\widetilde{R}_{r,t+1} \leq \alpha_2 \widetilde{R}_{r,0} \quad \text{and} \quad \widetilde{R}_{r,t+1} > \widetilde{R}_{r,t};$$

3. Excessively long inner loop:

$$t \geq \alpha_3 k;$$

where $\alpha_1 \in (0, \alpha_2), \alpha_2 \in (0,1)$, and $\alpha_3 \in (0,1)$ are user-defined parameters. Whenever any of the above conditions is satisfied, we terminate the current inner loop and begin a new outer iteration by setting $s^{r+1,0} = \bar{s}^{r,\tau_r}$ and updating the penalty parameter ρ_{r+1} accordingly.

The update rule for the penalty parameter ρ is also motivated by the complexity results in Theorem 1. Specifically, at the beginning of the $(r+1)$-th outer iteration, we determine ρ_{r+1} by solving the following minimization problem:

$$\rho_{r+1} = \arg\min_{\rho>0} \left\| s_Q^{r+1,0} - s^* \right\|_{\mathcal{H}}^2, \tag{60}$$

where s^* denotes a solution to the KKT system (59), and the metric $\left\| s_Q^{r+1,0} - s^* \right\|_{\mathcal{H}}$ corresponds to the upper bound derived in Theorem 1. The rationale behind this formulation is that minimizing this bound is expected to yield a smaller quantity $\left\| s_Q^{r+1,t} - \bar{s}_Q^{r+1,t+1} \right\|_{\mathcal{H}}$ for all $t \geq 0$, which, in turn, reduces the KKT residual $\left\| \mathcal{K}\left(\bar{s}_Q^{r+1,t+1} \right) \right\|$. This insight guides the dynamic adjustment of ρ to enhance the practical performance of the method.

Substituting the expressions of operator $\mathcal{H}$ and operator $\mathcal{O}_y$ into (60), the update rule for ρ_{r+1} reduces to solving the following one-dimensional minimization problem:

$$\rho_{r+1} = \arg\min_{\rho>0} \left\{ f(\rho) := \theta_1 \rho + \frac{\theta_2}{\rho} + \rho^2 \sum_{i=1}^{n} \frac{\theta_{3,i}}{1+\rho Q_{ii}} \right\}, \tag{61}$$

where the coefficients $\theta_1, \theta_2, \theta_3$ are given by

$$\begin{aligned}
\theta_1 &= \lambda_{AA^*} \left\| y^{r+1,0} - y^* \right\|^2 + \left\| w_Q^{r+1,0} - w^* \right\|_{Q^2}^2 \\
&\quad - 2 \left\langle Q\left(w_Q^{r+1,0} - w^* \right), A^*\left(y^{r+1,0} - y^* \right) \right\rangle, \\
\theta_2 &= \left\| x^{r+1,0} - x^* \right\|^2, \\
\theta_{3,i} &= Q_{ii} \left(A^* y^{r+1,0} - A^* y^* \right)_i^2, \quad i = 1, \ldots, n.
\end{aligned}$$

When coefficients $\theta_j > 0, j = 2,3$, the function $f(\rho)$ is strictly convex on $\rho > 0$, since its second derivative satisfies

$$f''(\rho) = \frac{2\theta_2}{\rho^3} + 2 \sum_{i=1}^{n} \frac{\theta_{3,i}}{(1+\rho Q_{ii})^3} > 0 \text{ for all } \rho > 0.$$

This guarantees both the existence and uniqueness of the optimal solution to problem (61). This scalar optimization problem can be efficiently solved using the golden section search method [4]. Moreover, since the exact values of θ_i involve the unknown solution s^*, we approximate them in practice as follows:

$$\begin{aligned}
\widetilde{\theta}_1 &= \lambda_{AA^*} \left\| \bar{y}^{r,\tau_r} - y^{r,0} \right\|^2 + \left\| \bar{w}_Q^{r,\tau_r} - w_Q^{r,0} \right\|_{Q^2}^2 \\
&\quad - 2 \left\langle Q\left(\bar{w}_Q^{r,\tau_r} - w_Q^{r,0} \right), A^*\left(\bar{y}^{r,\tau_r} - y^{r,0} \right) \right\rangle, \\
\widetilde{\theta}_2 &= \left\| \bar{x}^{r,\tau_r} - x^{r,0} \right\|^2, \\
\widetilde{\theta}_{3,i} &= Q_{ii} \left(A^* \bar{y}^{r,\tau_r} - A^* y^{r,0} \right)_i^2, \quad i = 1, \ldots, n.
\end{aligned} \tag{62}$$

Using these approximations, the updated penalty parameter ρ is obtained by solving the following scalar optimization problem:

$$\rho_{\text{new}} = \arg\min_{\rho>0} \left\{ f(\rho) = \widetilde{\theta}_1 \rho + \frac{\widetilde{\theta}_2}{\rho} + +\rho^2 \sum_{i=1}^{n} \frac{\widetilde{\theta}_{3,i}}{1+\rho Q_{ii}} \right\}. \tag{63}$$

To further stabilize the update and prevent the approximations from deviating significantly from the true values, we apply an exponential smoothing scheme. The complete update procedure for ρ is summarized in Algorithm 1:

Algorithm 1 PenaltyUpdate procedure

1: Input: $\bar{s}_Q^{r,\tau_r}$, $s_Q^{r,0}$, $\mathcal{O}_y$, A, and Q.
2: Compute $\widetilde{\theta}_1$, $\widetilde{\theta}_2$, and $\widetilde{\theta}_3$ as defined in (62);
3: Ensure numerical stability:

$$\widetilde{\theta}_i \leftarrow \max\left(\widetilde{\theta}_i, 10^{-12} \right), \quad i = 1, 2;$$

4: Solve problem (63) via the golden section search method to obtain ρ_{new};
5: Compute the smoothing factor:

$$\beta = \exp\left(-\widetilde{R}_{r,\tau_r - 1} / \widetilde{R}_{0,\tau_0 - 1} \right);$$

6: Update ρ by

$$\rho_{r+1} = \exp\left(\beta \log(\rho_{\text{new}}) + (1-\beta) \log(\rho_r) \right);$$

7: Output: ρ_{r+1}.

Appendix D: The sGS-dHPR-ED algorithm with Restart Strategy and ρ Updating Scheme

Algorithm 2 A sGS-dHPR-ED algorithm for solving the SCED problem with restart strategy and ρ updating scheme

1: Input: Let $\mathcal{O}_y$ be a self-adjoint positive semidefinite linear operator on $\mathbb{R}^m$ such that $\mathcal{O}_y + AA^*$ is positive definite. Denote $s_Q = (y, w_Q, z, x)$ and $\bar{s}_Q = (\bar{y}, \bar{w}_Q, \bar{z}, \bar{x})$. Let $s_Q^0 = (y^0, w_Q^0, z^0, x^0) \in \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n$.
2: Initialization: Set outer loop counter $r = 0$, total iteration counter $k = 0$, and initial penalty parameter $\rho > 0$.
3: **repeat**
4: Initialize inner loop: set inner counter $t = 0$;
5: **repeat**
6: $\bar{z}^{r,t+1} = \arg\min_{z \in \mathbb{R}^n} L_{\rho_r}\left(y^{r,t}, w_Q^{r,t}, z; x^{r,t} \right)$;
7: $\bar{x}^{r,t+1} = x^{r,t} + \rho_r \left(-Q w_Q^{r,t} + A^* y^{r,t} + \bar{z}^{r,t+1} - c \right)$;
8: $\bar{w}_Q^{r,t+\frac{1}{2}} = D_{Q,r}\left(\rho_r Q w_Q^{r,t} + 2\bar{x}^{r,t+1} - x^{r,t} \right)$; where $D_{Q,r} = \operatorname{diag}\left(\frac{1}{1+\rho_r Q_{ii}} \right), i = 1, \ldots, n$ is a diagonal matrix.
9: $\bar{y}^{r,t+1} = \arg\min_{y \in \mathbb{R}^m} \left\{ L_{\rho_r}\left(y, \bar{w}_Q^{r,t+\frac{1}{2}}, \bar{z}^{r,t+1}; \bar{x}^{r,t+1} \right) + \frac{\rho_r}{2} \left\| y - y^{r,t} \right\|_{\mathcal{O}_y}^2 \right\}$.
10: $\bar{w}_Q^{r,t+1} = \bar{w}_Q^{r,t+\frac{1}{2}} + \rho_r D_{Q,r} \cdot A^*(\bar{y}^{r,t+1} - y^{r,t})$;
11: $\hat{s}_Q^{r,t+1} = 2\bar{s}_Q^{r,t+1} - s_Q^{r,t}$;
12: $s_Q^{r,t+1} = \frac{1}{k+2} s_Q^{r,0} + \frac{k+1}{k+2} \hat{s}_Q^{r,t+1}$;
13: $t \leftarrow t+1, \quad k \leftarrow k+1$;
14: **until** restart or termination criteria are met
15: Restart: Set $\tau_r = t$, $s_Q^{r+1,0} = \bar{s}_Q^{r,\tau_r}$;
16: $\rho_{r+1} = \text{PenaltyUpdate}\left(\bar{s}_Q^{r,\tau_r}, s_Q^{r,0}, \mathcal{O}_y, A, Q \right)$;
17: $r \leftarrow r+1$;
18: **until** termination criteria are met
19: Output: $\left\{ \bar{s}^{r,t} \right\}$.

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